Reviewer Pack — Kahler Manifold Rank to DPLL Search Tree Height Ratio

Entry 5adb788e9c18 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Kahler Manifold Rank to DPLL Search Tree Height Ratio

Entry ID: `5adb788e9c18` **Verdict**: `INCONCLUSIVE` **Recorded**: 2026-05-28 23:48:05 UTC

1. Conjecture

Field A (mathematical branch): Algebraic Geometry (Kähler Manifolds) **Field B** (complexity object): Complexity Theory: DPLL Search Tree Complexity

Statement:

> For every satisfiable 3-CNF formula F with n variables, the ratio of the rank of its associated Kähler manifold to the height of the corresponding DPLL search tree is at most a constant c.

Rationale (proposer's reasoning):

> Kähler manifolds provide geometric structures related to polynomials that are believed to be hard to classify. If the rank of such a structure can be bounded by a polynomial in the size of the formula, it may imply an efficient algorithm for solving the formula or conversely reveal difficulty in the DPLL search process.

Taxonomy category: `KAHLER_MANIFOLD_RANK_TO_DPLL_HEIGHT_RATIO` (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): `4266c4b8a2140071`

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

> For a satisfiable 3-CNF formula F with n variables, if the ratio of its Kähler manifold rank to DPLL search tree height exceeds a constant c for any seed or instance, the conjecture is falsified.

3. Barrier filter (F1)

Final: `PASS`

Barrier	Verdict	Confidence	LLM_1	LLM_2
RELATIVIZATION	UNCERTAIN	1.00	HITS	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN

Barrier	Verdict	Confidence	LLM₁	LLM₂
ALGEBRIZATION	SAFE	0.80	SAFE	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "Kähler manifold rank" AND "DPLL search tree height" - "algebraic geometry" AND "complexity theory" AND DPLL - "satisfiable 3-CNF formula" AND Kähler manifold AND DPLL search tree

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)
Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution**: rc=1, elapsed=0.1s

5.1 Generated Python source

```python
import random
import math
from fractions import Fraction
from itertools import combinations, chain

def generate_3cnf(n: int) -> list:
    variables = [f'x{i}' for i in range(1, n+1)]
    clauses = []
    for _ in range(n):
        clause = random.sample(variables + [f'~{v}' for v in variables], 3)
        clauses.append(clause)
    return clauses

def is_satisfiable(clauses: list) -> bool:
    def dpll(clauses, assignment):
        if not clauses:
            return True
        unit_clause = next((c for c in clauses if len(c) == 1), None)
        if unit_clause:
            var = unit_clause[0]
            value = var.startswith('~') ^ (var[1:] not in assignment)
            new_assignment = {**assignment, var[1:]: value}
            return dpll([c for c in clauses if var not in c and '~' + var not in c], new_assignmer
        pure_literal = next((v for v in variables if all(v not in c or '~' + v in c for c in claus
        if pure_literal:
            value = pure_literal.startswith('~') ^ (pure_literal[1:] not in assignment)
            new_assignment = {**assignment, pure_literal[1:]: value}
            return dpll(clauses, new_assignment)
        p_var = random.choice(variables)
        return dpll(clauses + [[f'~{p_var}']], {**assignment, p_var: True}) or dpll(clauses + [[p_
    return dpll(clauses, {})

def kahler_manifold_rank(formula: list) -> int:
    # Placeholder for Kähler manifold rank calculation
    # This is a dummy implementation and should be replaced with actual computation
```

```
    return len(formula)

def dpll_search_tree_height(clauses: list) -> int:
    def dpll(clauses, assignment):
        if not clauses:
            return 0
        unit_clause = next((c for c in clauses if len(c) == 1), None)
        if unit_clause:
            var = unit_clause[0]
            value = var.startswith('~') ^ (var[1:] not in assignment)
            new_assignment = {**assignment, var[1:]: value}
            return dpll([c for c in clauses if var not in c and '~' + var not in c], new_assignmer
        pure_literal = next((v for v in variables if all(v not in c or '~' + v in c for c in claus
        if pure_literal:
            value = pure_literal.startswith('~') ^ (pure_literal[1:] not in assignment)
            new_assignment = {**assignment, pure_literal[1:]: value}
            return dpll(clauses, new_assignment) + 1
        p_var = random.choice(variables)
        return max(dpll(clauses + [[f'~{p_var}']], {**assignment, p_var: True}), dpll(clauses + [[
    return dpll(clauses, {})

def run_trial(seed: int) -> dict:
    random.seed(seed)
    n = random.randint(5, 40)
    formula = generate_3cnf(n)
    if not is_satisfiable(formula):
        return {
            "metric_name": "Kahler Manifold Rank to DPLL Search Tree Height Ratio",
            "metric_value": None,
            "instances_tested": 1,
            "conjecture_holds": False,
            "counterexample": "unsatisfiable_formula"
        }
    rank = kahler_manifold_rank(formula)
    height = dpll_search_tree_height(formula)
    ratio = Fraction(rank, height) if height != 0 else None
    return {
        "metric_name": "Kahler Manifold Rank to DPLL Search Tree Height Ratio",
        "metric_value": float(ratio) if ratio is not None else None,
        "instances_tested": 1,
        "conjecture_holds": True,
        "counterexample": ""
    }

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [random.randint(2, 97) for _ in range(30)]
    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    if all(r["conjecture_holds"] for r in results):
        mean_value = sum(r["metric_value"] for r in results) / len(results)
        support_fraction = 1.0
```

```
        result = f"RESULT: SUPPORTED mean={mean_value} std=0.0 support_fraction={support_fraction}
    else:
        first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
        counterexample = "unknown"
        result = f"RESULT: FALSIFIED counterexample=\"{counterexample}\" first_failing_seed={first

    print(result)
```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ca80751b.py", line 96, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ca80751b.py", line 72, in run_trial
    if not is_satisfiable(formula):
           ^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ca80751b.py", line 42, in is_satisfiabl
    return dpll(clauses, {})
           ^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ca80751b.py", line 35, in dpll
    pure_literal = next((v for v in variables if all(v not in c or '~' + v in c for c in clauses)), None)
                                                 ^^^^^^^^^
NameError: name 'variables' is not defined
```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

> skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

> The test code crashed before producing data, which means we cannot verify the conjecture's support or falsification. | next: Investigate and fix the crash in the test code to proceed with the verification of the conjecture.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	10077
2	preregistration	ollama_remote	glm4:latest	0	0	5692
3	novelty	ollama_remote	glm4:latest	0	0	4726

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
4	novelty	ollama_remote	glm4:latest	0	0	6167
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	16762
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10069
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10803
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	13812
9	judge	ollama_remote	glm4:latest	0	0	14652

Totals: 0 input tokens, 0 output tokens, ~$0.0000 cost, 92759 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/5adb788e9c18.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/5adb788e9c18.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)
Replay tarball: `research/replay/5adb788e9c18.tar.gz` (if generated)